

Population organization and closed-loop attractors for obstacle avoidance under perturbation

Summary

Motor systems must adjust ongoing actions after unexpected forces while maintaining goal-directed movement within spatial boundaries. This resembles biological reaching: unexpected forces prompt graded corrections, supported by coordinated muscle activity, while the hand continues toward its goal. How can a learned controller organize such corrections when obstacles narrow the available route? We asked whether reward-based training alone can produce structured obstacle avoidance and whether its activity offers an organizational comparison with biological motor control. A proximal policy optimization (PPO) actor-critic was trained for 3 million environment steps. Its feedforward multilayer perceptron (MLP), with two 256-unit hidden layers, mapped each state to an action moving a point mass through a corridor with two central obstacles. We analyzed hidden-unit activity using principal component analysis, clustering, kinematic tuning, linear decoding, and closed-loop analysis of the agent and environment physics. Moderate lateral perturbations changed the path by under 0.1 m without stopping progress toward the target. The first two principal components captured 71% of population variance; activity followed phase-ordered trajectories shared across perturbation conditions and separated by detour route. At the obstacle band, closed-loop analysis identified a stable lateral fixed point for each route ($x^*=3.32$ m and 6.71 m), with similar relaxation times (1.42 s and 1.34 s) and clearances (0.43 m and 0.46 m). In this model, reward training produced structured routes with local stability. This offers a preliminary point of comparison with biological motor control; the model is not a neural system, and results come from one training run.

Additional details

To understand how the controller produced these routes, we looked at both its activity and its interaction with the task. The term "population" refers to the 256 artificial units in one hidden layer, not neurons recorded from an animal. The feedforward actor has no recurrent memory: it maps the current observation to an action. The fixed points are therefore not equilibria of the hidden layer alone; they describe the policy acting together with the point-mass physics. This differs from recurrent-network fixed-point analyses (Sussillo & Barak, 2013). Motor-cortex population studies provide a useful organizational comparison (Churchland et al., 2012), but artificial units are not neurons.

The controller was trained for 3 million environment steps. We evaluated a baseline and six conditions with lateral force impulses in three magnitudes to the left or right, each lasting 14 timesteps. Our analysis combined principal component analysis, clustering, kinematic tuning, linear decoding, and a closed-loop analysis of the policy with the environment physics.

Unexpected forces can also prompt graded corrections during human reaching, with coordinated muscle activity while the hand continues toward its goal (Pruszynski & Scott, 2012; Nashed et al., 2014). The resemblance here is organizational only; it does not show that the model uses the same neural or muscular mechanisms. These results come from one training run, and fixed-point locations change with the checkpoint. Further runs and direct comparisons with human movement will be needed to see how robust or biologically informative this account is.

How the controller and environment form a closed loop

At each step, the feedforward policy uses the current observation, including position, velocity, goal, and obstacle locations, to choose an action. The environment applies that action together with any external force and updates the point mass. We analyze the behavior of this combined policy-and-physics system, rather than treating the hidden layer as if it had memory of its own.

At the obstacle band ($y = 5$ m), we focused on lateral position and velocity (x and v_x), with forward speed held at its plateau and the perturbation turned off. For each route, a fixed point is where the commanded lateral acceleration is zero. Its negative slope means that small deviations to either side elicit a correction back toward the route. We also checked the local dynamics: real eigenvalues with magnitudes below one indicate nearby states return without oscillating.

We found one stable, non-oscillatory fixed point for each route. Their positions (3.32 m and 6.71 m) were within 0.11 m and 0.09 m of where successful episodes crossed the obstacle band; clearances were 0.43 m and 0.46 m. The 1.42 s and 1.34 s relaxation times describe the model's local return toward those points, not human response times. Figure 1 brings the trajectories, activity, and lateral dynamics together.

We also tested whether the hidden activity contained information about the perturbation. A linear decoder, evaluated on held-out episodes, identified the condition with 0.81 accuracy after the impulse (chance = 0.14), but only 0.11 before it. Most errors were between neighboring force magnitudes on the same side. This shows that perturbation information could be read from activity after the force; it does not show that the policy used that information to choose its action (Figure 2). Taken together, the results suggest that reward training can produce organized, locally stable routes in this model. They are an initial comparison point for biological motor control, not evidence that the model reproduces neural control.

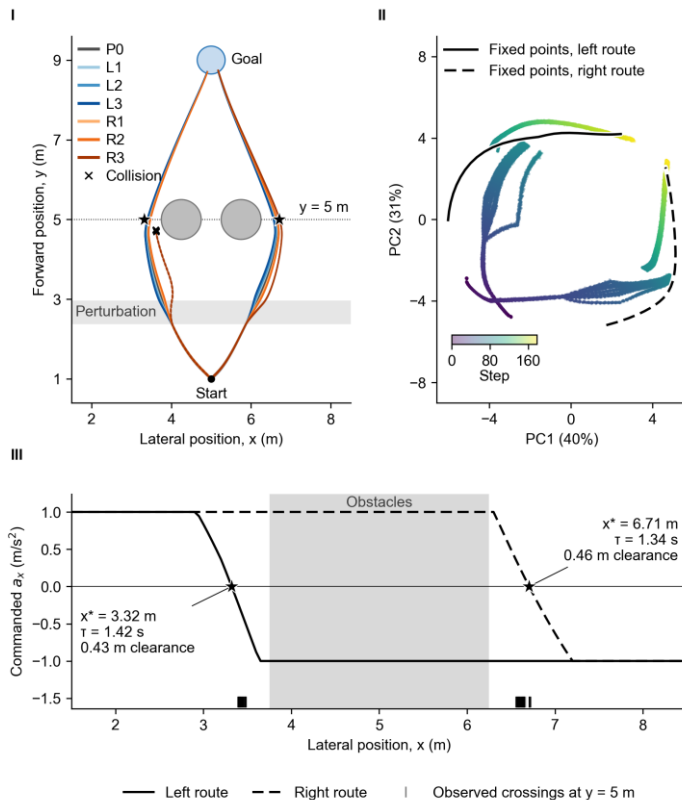


Figure 1. Task and closed-loop results. (A) Baseline and perturbed trajectories; only the strongest rightward impulse caused collisions, on the left route. Stars mark route fixed points at $y = 5$ m. (B) Hidden activity in PC1-PC2 space, colored by step. (C) Commanded lateral acceleration; each route has a stable, negative-slope zero crossing.

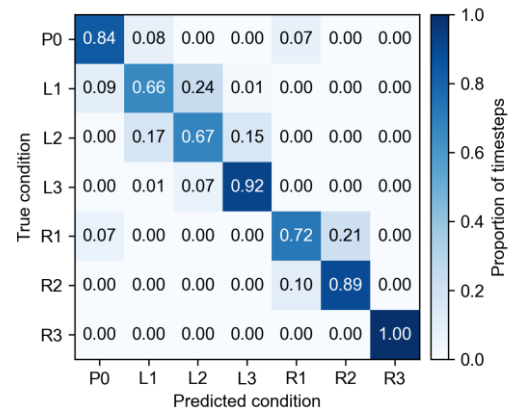


Figure 2. Prediction of perturbation condition from hidden activity after the impulse. Rows show true conditions and columns predictions; values are row-normalized proportions from episode-held-out validation (accuracy = 0.81; chance = 0.14). Most errors are between neighboring magnitudes on the same side.